Math for Economists Bootcamp

Module I: Linear Algebra, Part 1

Terrie Walmsley
Department of Economics
USCDornsife

Vectors and Matrices

- A way of arranging data from a table
  - $A = \begin{bmatrix} 10 & 20 & 10 \\ 15 & 15 & 15 \\ 11 & 12 & 15 \\ 21 & 15 & 10 \end{bmatrix}$

4x3 matrix

Firm	Number of workers (L)	Capital (K)	Profit (P)
1	10	20	10
2	15	15	15
3	11	12	15
4	21	15	10

- In general form:
  - A is an mxn matrix
  - $A = \begin{bmatrix} a11 & a1j & a1n \\ \dots & aij & \dots \\ am1 & \dots & amn \end{bmatrix}$
  - Where m is number of rows and n is the number of columns

Special Matrices

- Square (mxn where m=n)
- Triangular (one side of diagonal is 0), e.g., $\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$ or $\begin{bmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{bmatrix}$
- diagonals (only diagonal are non-zero), e.g., $\begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & f \end{bmatrix}$
- identity (I) (diagonal, where diagonal is one), e.g., $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- zero/null (all zeros), e.g., $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Matrix Operations and Rules

- Matrices that have the same dimensions are conformable for addition. If A and B have the same dimension, then: $A \pm B = [a_{ij} \pm b_{ij}]$
- Matrices are multiplied by **inner product** of row by column. Matrices are conformable for multiplication if they have the same inner dimensions:
  - $\underset{(mxn)}{\mathrm{A}} * \underset{(nxk)}{\mathrm{B}} = \underset{(mxk)}{\mathrm{C}}$
  - $\underset{(nxm)}{\mathrm{A}} * \underset{(nxk)}{\mathrm{B}} = not\ possible$
- Suppose A, B and C are conformable matrices for either addition or multiplication, then:
  - $(A + B) + C = A + (B + C)$
  - $(AB)C = A(BC)$
  - $A * (B + C) = AB + AC$

Operations

- Adding/subtracting
  - $$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \\ a_{31}+b_{31} & a_{32}+b_{32} & a_{33}+b_{33} \end{bmatrix}$$
- Transpose, for all entries: $a_{ij} \to a_{ji}$
  - $$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$
  - Symmetric matrix (square matrix is equal to its transpose i.e., $a_{ij} = a_{ji}$)

Examples

- $$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 4 & 2 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 7 \\ 6 & 7 \\ 9 & 11 \end{bmatrix}$$

Examples

- $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 4 & 2 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 7 \\ 6 & 7 \\ 9 & 11 \end{bmatrix}$

- $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 2 & 0 \end{bmatrix}$

Examples

- $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$
- *3x2* *2x3*

Multiplication

- By scalar

$$-\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} * \lambda = \begin{bmatrix} \lambda * a_{11} & \lambda * a_{12} & \lambda * a_{13} \\ \lambda * a_{21} & \lambda * a_{22} & \lambda * a_{23} \\ \lambda * a_{31} & \lambda * a_{32} & \lambda * a_{33} \end{bmatrix}$$

Examples

- $2 * \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 8 \\ 4 & 10 \\ 6 & 12 \end{bmatrix}$

Multiplication

- Two matrices
  - $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} * \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$

$$= \begin{bmatrix} a_{11} * b_{11} + a_{12} * b_{21} + a_{13} * b_{31} & a_{11} * b_{12} + a_{12} * b_{22} + a_{13} * b_{32} & a_{11} * b_{13} + a_{12} * b_{23} + a_{13} * b_{33} \\ a_{21} * b_{11} + a_{22} * b_{21} + a_{23} * b_{31} & a_{21} * b_{22} + a_{22} * b_{22} + a_{23} * b_{32} & a_{21} * b_{13} + a_{22} * b_{23} + a_{23} * b_{33} \\ a_{31} * b_{11} + a_{32} * b_{21} + a_{33} * b_{31} & a_{31} * b_{32} + a_{32} * b_{22} + a_{33} * b_{32} & a_{31} * b_{13} + a_{32} * b_{23} + a_{33} * b_{33} \end{bmatrix}$$

- Note:
  - Columns of matrix 1 must equal rows of matrix 2
  - Resulting matrix has dimensions = rows of matrix 1 x columns of matrix 2

Examples

- $\begin{bmatrix} 1 & 2 \\ 3 & 2 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$
  - Not possible (3x2) x (3x3) doesn't match
  - But you could transpose $\begin{bmatrix} 1 & 2 \\ 3 & 2 \\ 2 & 1 \end{bmatrix}$ and then multiply
- $\begin{bmatrix} 1 & 3 & 2 \\ 2 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1+12+14 & 2+15+16 & 3+18+18 \\ 2+8+7 & 4+10+8 & 6+12+9 \end{bmatrix} = \begin{bmatrix} 27 & 33 & 39 \\ 17 & 22 & 27 \end{bmatrix}$
  - (2x3) x (3x3) and solution is 2x3

Vectors and Matrices

- Vectors and matrices are a way of organizing a system of equations:
  - $a11 * x_1 + a12 * x_2 = a$
  - $a21 * x_1 + a22 * x_2 = b$
- In matrix format:
  - In matrix notation: $A * \boldsymbol{x} = \boldsymbol{c}$
  - $\begin{bmatrix} a11 & a12 \\ a21 & a22 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \mathrm{a} \\ b \end{bmatrix}$
    - Where:
      - A is a 2x2 matrix with the coefficients as entries (note use of upper case A for matrices)
      - **x** and **c** are column vectors of dimension 2 (2x1) (note use of lower case and bold for vectors)

Solve

System of Equations

$x + y + 2z = 1$
$x + 2y + 2z = 2$
$y + z = 3$

- $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ where $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

System of Equations

$x + 3y + 2z = 1$
$x + 2y + 2z = 2$
$y = 3$

- $\begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix} . \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ where $A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix}$

Gauss-Jordan method

- Method for solving a set of simultaneous equations
- Aim is to reduce the matrix into row echelon form:
  - Triangular matrix with leading 1's along the diagonal.
  - E.g., $\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$ or $\begin{bmatrix}1 & 0 & 2\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$
- How is this done?
  - Interchange rows
  - Add/subtract multiples of other rows: $R_i \Rightarrow a * Rj + b * Ri$
  - Until matrix is in row echelon form

Solve

System of Equations

$$x + y + 2z = 1$$
$$x + 2y + 2z = 2$$
$$y + z = 3$$

- $\begin{bmatrix}1 & 1 & 2\\1 & 2 & 2\\0 & 1 & 1\end{bmatrix} . \begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}$ where $A = \begin{bmatrix}1 & 1 & 2\\1 & 2 & 2\\0 & 1 & 1\end{bmatrix}$

Solve

Identity matrix $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

- $\begin{bmatrix}1&1&2\\1&2&2\\0&1&1\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix}$
  - R_2=-R_1+R_2 → $\begin{bmatrix}1&1&2\\0&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1\\1\\3\end{bmatrix}$
  - R_1= -R_2+R_1 → $\begin{bmatrix}1&0&2\\0&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}0\\1\\3\end{bmatrix}$
  - R_3=-R_2+R_3 → $\begin{bmatrix}1&0&2\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}0\\1\\2\end{bmatrix}$

R_2=-R_1+R_2

$A_{21} \Rightarrow -(1) + 1 = 0$

$A_{22} \Rightarrow -(1) + 2 = 1$

$A_{23} \Rightarrow -(2) + 2 = 0$

$V_2 \Rightarrow -(1) + 2 = 1$

R_1=-R_2+R_1

$A_{11} \Rightarrow -(0) + 1 = 1$

$A_{12} \Rightarrow -(1) + 1 = 0$

$A_{13} \Rightarrow -(0) + 2 = 2$

$V_1 \Rightarrow -(1) + 1 = 0$

Solve

Identity matrix $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

- $\begin{bmatrix}1&1&2\\1&2&2\\0&1&1\end{bmatrix}.\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\2\\3\end{bmatrix}$ or $\begin{bmatrix}1&1&2\\1&2&2\\0&1&1\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix}$
  - R_2=-R_1+R_2 → $\begin{bmatrix}1&1&2\\0&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1\\1\\3\end{bmatrix}$
  - R_1= -R_2+R_1 → $\begin{bmatrix}1&0&2\\0&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}0\\1\\3\end{bmatrix}$
  - R_3=-R_2+R_3 → $\begin{bmatrix}1&0&2\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}0\\1\\2\end{bmatrix}$
  - R_1=-2.R_3+R_1 → $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}-4\\1\\2\end{bmatrix}=\begin{bmatrix}x\\y\\z\end{bmatrix}$
  - This is row echelon form since leading ones in each row with zeros in their columns
  - x=-4, y=1 and z=2
  - A is non-Singular and is solve-able

Gauss-Jordan method

- A matrix must be non-singular, which makes it invertible, for it to be solve-able.
- Options for solutions
  - Unique, none and multiple
  - Also, number of equations must equal number of unknowns.
    - Overdetermined – more equations than variables (may not be a solution)
    - Under-determined – less equations than variables. Must set some of the under-determined variables as exogenous, so other (endogenous) variables become functions of exogenous variable.
- A good video explaining the Guass-Jordan method is available at: https://www.youtube.com/watch?v=eDb6iugi6Uk

Solve

System of Equations

$$x + y + 2z = 1$$
$$x + 2y + 2z = 2$$
$$y + z = 3$$

- $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ where $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

System of Equations

$$x + 3y + 2z = 1$$
$$x + 2y + 2z = 2$$
$$y = 3$$

- $\begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix} . \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ where $A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 0 \end{bmatrix}$

Solve

- $\begin{bmatrix}1 & 1 & 2\\1 & 2 & 2\\0 & 1 & 0\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix}$
  - Swap R3 and R2 $\begin{bmatrix}1 & 1 & 2\\0 & 1 & 0\\1 & 2 & 2\end{bmatrix}\begin{bmatrix}1\\3\\2\end{bmatrix}$
  - R_3=-R_1+R_3 → $\begin{bmatrix}1 & 1 & 2\\0 & 1 & 0\\0 & 1 & 0\end{bmatrix}\begin{bmatrix}1\\3\\1\end{bmatrix}$
  - Looks like we have an issue y=1 and y=3 So let's expand the matrix and go back to the equations:
    - $\begin{bmatrix}1 & 1 & 2\\0 & 1 & 0\\0 & 1 & 0\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\3\\1\end{bmatrix}$
    - Which gives us: $x + y + 2z = 1$, y = 3, and y = 1
    - There are two solutions for y and insufficient equations to determine x and z independently.
  - In some cases we could exogenize (fix) either x or z; and then make the other a function of the exogenous variable (e.g., $x = -2 * z_0$), but in this case there are no solutions that satisfy all three equations.
  - A is Singular (i.e, there is no solution)

R_3=-R_1+R_3
A_{31} -> -1+1=0
A_{32} -> -1+2=1
A_{33} -> -2+2=0

V_3 -> -1+2=1

Operations

- There is no division in linear algebra.
- To do division we need to find the inverse (A^{-1}) of a matrix.
- That is the matrix, which when multiplied with the original matrix, gives us the identity matrix.

$$(\mathrm{A} * \mathrm{A}^{-1}) = \mathrm{I}$$

- So for:
  - $\mathrm{A} * \mathbf{x} = \mathbf{y}$
  - $\mathrm{A} * \mathrm{A}^{-1}\mathbf{x} = \mathrm{A}^{-1}\mathbf{y}$
  - $\mathbf{I} * \mathbf{x} = \mathrm{A}^{-1}\mathbf{y}$
  - $\mathbf{x} = \mathrm{A}^{-1}\mathbf{y}$

To determine A^{-1}, you first need to find the determinant

- Determinant of a 2x2
  - $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
  - $|A| = ad - bc$
- Determinant of an nxn (3x3)
  - $A = \begin{bmatrix} a11 & a12 & a13 \\ a21 & a22 & a23 \\ a31 & a32 & a33 \end{bmatrix}$
    - Note that minors of a matrix: e,g., $M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = a_{21} * a_{33} - a_{23} * a_{31}$ is the determinant of the matrix with row 1 and column 2 removed
    - And the Co-factors of A: $C_{ij} = (-1)^{i+j}.\, M_{ij}$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

  - To obtain the determinant $|A|$, you then sum the weighted co-factors for one column or row:
    - $|A| = \sum_j^m a_{ij}.C_{ij}$ for any i, or
    - $|A| = \sum_i^n a_{ij}.C_{ij}$ for any j

$|A| = \det$

Some other things about determinants

- $|A| = |A^T|$
- $|AB| = |A| * |B|$
- If two rows/columns are identical then $|A| = 0$
- If B is obtained from A by
  - interchanging rows – then $|B| = -|A|$
  - taking R_i -> $a*R_j + b*R_i$, then $|B| = |A|$
  - multiplying a row by constant c – then $|B| = c|A|$
  - multiplying all (2) rows by c – then $|B| = c * c * |A|$

Example: find the determinant

- $A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

Using column 1

– $j = 1\ (|A| = \sum_j^m a_{1j} * C_{1j})$
  - $1 \times (2 \times 1 - 2 \times 1) - 1 \times (3 \times 1 - 2 \times 1) + 0 \times (3 \times 2 - 2 \times 2)$
  - $1 \times (0) - 1 \times (1) + 0 \times (2)\ = -1$

- $|A| = \sum_j^m a_{ij} * C_{ij}$ no matter what i you choose (can also do by columns)
  – $i = 1\ (|A| = \sum_j^m a_{1j}.C_{1j})$
    - $1 \times (0) + 3 \times (-1) + 2 \times (1) = -1$

  – $i = 2\ (|A| = \sum_j^m a_{2j} * C_{2j})$
    - $0\ \times (2) + 1 \times (0) + 1 \times (-1) = -1$

  – $i = 3\ (|A| = \sum_j^m a_{3j} * C_{3j})$
    - $1 \times -(1) + 2 \times (1) + 2 \times -(1) = -1$

Inverse of a matrix

- For a 2x2:
  – If A = $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$
  – $A^{-1} = \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \times (\frac{1}{|A|})$
    - Where $|A|$ is the determinant
- For an nxn matrix:
  – $A^{-1} = \begin{bmatrix} C_{11} & \dots & C_{1n} \\ \dots & \dots & \dots \\ C_{n1} & \dots & C_{nn} \end{bmatrix}^T \times (\frac{1}{|A|})$
  – Where C_{ij} are the co-factors, $C_{ij} = (-1)^{i+j}.\ M_{ij}$
  – Where M_{ij} are the minors of a matrix. That is the determinant of the matrix with row i and column j removed

Example: Find the inverse

- $A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}$
- Known from previous exercise: $|A| = -1$
- $A^{-1} = \begin{bmatrix} C_{11} & \dots & C_{1n} \\ \dots & \dots & \dots \\ C_{n1} & \dots & C_{nn} \end{bmatrix}^T \times (\frac{1}{|A|})$
- $A^{-1} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 1 & -1 \\ 2 & 0 & -1 \end{bmatrix}^T \times \left(\frac{1}{-1}\right)$

$$= \begin{bmatrix} 0 & -1 & 2 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix} \times \left(\frac{1}{-1}\right) = \begin{bmatrix} 0 & 1 & -2 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

- Check: $A * A^{-1} = I$
- $\begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix} . \begin{bmatrix} 0 & 1 & -2 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

27

University of Southern California

Inverse Matrix Rules and Things to Note

- A square matrix is invertible if and only if its determinant is non-zero
  - $A^{-1} exists \iff |A| \neq 0$

- $(A^{-1})^{-1} = A$
- $(A^T)^{-1} = (A^{-1})^T$
- $(AB)^{-1} = B^{-1}A^{-1}$

28

University of Southern California

Solving a System of linear equations using Cramer's rule

- Any system of equations can be summarized in the form
  - $\mathrm{A} * \mathbf{x} = \mathbf{b}$
  - Where A is an nxn matrix of coefficients a_{ij}
  - x is a column vector of n variables
  - b is a column vector of n values
  - Crammers rule for solving **x**

$$X_j = \frac{|A_j|}{|A|}$$

- Where A_j is the A matrix with the jth column replaced with the vector **b**

Cramer's Rule

- $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$
  - $|A| = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 1 \end{vmatrix} = 1*(2-2) - 1*(1-0) + 2*(1-0) = 1$
  - $|A_1| = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 2 & 2 \\ 3 & 1 & 1 \end{vmatrix} = 1*(2-2) - 1*(2-6) + 2*(2-6) = -4$
  - $|A_2| = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 3 & 1 \end{vmatrix} = 1*(2-6) - 1*(1-0) + 2*(3-0) = 1$
  - $|A_3| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 1 & 3 \end{vmatrix} = 1*(6-2) - 1*(3-0) + 1*(1-0) = 2$

- *Check:* $x = -4$, $y = 1$, and $z = 2$ is a solution
  - x+y+2z=1
    - =-4+1+2x2=1
  - x+2y+2z=2
    - =-4+2+4=2
  - y+z=3
    - =1+2=3